

IMPROVE INTERNET COVERAGE WITH MESH ROUTERS

This article lists the important purchase parameters along with some budget-friendly options for Wi-Fi based mesh routers that will help you choose the right one for your own use.

Today, even though we use technologies that once seemed futuristic, we all have struggled with slow Internet speeds and Wi-Fi dead zones at some point or the other. For large buildings, or those with many structural hindrances, traditional routers require additional components such as range extenders.

A mesh router offers easy setup with multiple satellite nodes (access points) controlled through a single panel that captures the router's signal, and rebroadcasts the same to create a strong and reliable Wi-Fi signal through a Single Service Set Identifier (SSID).

How to make the right choice

The common feature of all mesh systems is smart roaming. When the signal from the current node decreases, the connection gets switched to the one that offers the strongest connection. Some of the parameters that need to be considered to ensure that the router fits your requirements are discussed below.

Speed: One of the major advantages of a mesh router is speed when compared to a traditional router. Speed in the specifications is usually mentioned by combining the throughputs on individual frequency bands on which the router works. Internet plans that support these speeds are necessary to deliver the desired output.

Frequency spectrum: Wi-Fi systems usually operate on 2.4GHz and 5GHz frequency bands, and use 802.11ac wireless technology. A router may be dual- or tri-band capable to reduce congestion in the 2.4GHz spectrum.

Tri-band Wi-Fi technology establishes a dedicated backhaul connection between the router and satellite. The bands remain the same, but the 5GHz spectrum is split to create two separate networks. In both, wireless clients are connected to the optimal band based on network traffic conditions.

Coverage: The main reason for purchasing a mesh router is improved coverage. With there being the benefit of satellite nodes, you can achieve complete coverage in any home or building, no matter how big it is or how it is constructed.

Wired connectivity: Mesh systems usually come with two Ethernet ports for a direct and stronger Internet connection. Some of these are auto-configurable, and can act as both a wide area network (WAN) and local area network (LAN) based on whether these get connected to the gateway or the client.

Quality of service (QoS): This feature allows prioritisation of important tasks over less significant ones by identification of data packets over a network. This ensures that tasks

such as video streaming and calling happen without any interruption.

MU-MIMO: Some models support multi-user multiple input multiple output (MU-MIMO) technology, a feature in IEEE 802.11ac Wave 2, where data signals are sent and received from different devices simultaneously (in parallel) rather than sequentially to provide higher speed. Bandwidth gets evenly divided and the devices do not have to wait for long time periods to receive the signals. Multiplexed signals are termed as spatial streams. But for this feature to be functional, the client must also support MIMO.

Many routers are completely automated, with features such as automatic software updates and security checks. This makes remote management through a user-friendly mobile app or Web-based interface possible. Most apps allow users to perform functions like scanning network speeds, setting parental controls, managing guest networks, device prioritisation and more. This improves efficiency of the network.

Apart from this, some mesh systems may also be integrated with smart homes. This allows them to communicate with other devices that use the same protocols.

Market options

In this section we provide you with an overview of some options to help you decide which one may work for